

Architecture Design

HR ANALYTICS

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1. Introduction

1.1 What is Architecture design document?

Any software needs the architectural design to represent the design of software. IEEE defines architectural design as “the process of defining a collection of hardware and software components and their interfaces to establish the framework for the development of a computer system.” The software that is built for computer-based systems can exhibit one of these many architectures.

Each style will describe a system category that consists of :

- A set of components (eg: a database, computational modules) that will perform a function required by the system.
- The set of connectors will help in coordination, communication, and cooperation between the components.
- Conditions that how components can be integrated to form the system.
- Semantic models that help the designer to understand the overall properties of the system.

1.2 Scope

The scope of this Architecture Design Document is to define the architecture of the HR Analytics project. It will cover:

- The overall architecture design for the project.
- Detailed descriptions of key components, including data processing, analysis, and modeling.
- Deployment strategies for the machine learning models and analysis results.
- Communication flow within the architecture.

This document will serve as a reference for team members involved in the implementation and maintenance of the HR Analytics project.

2. Architecture

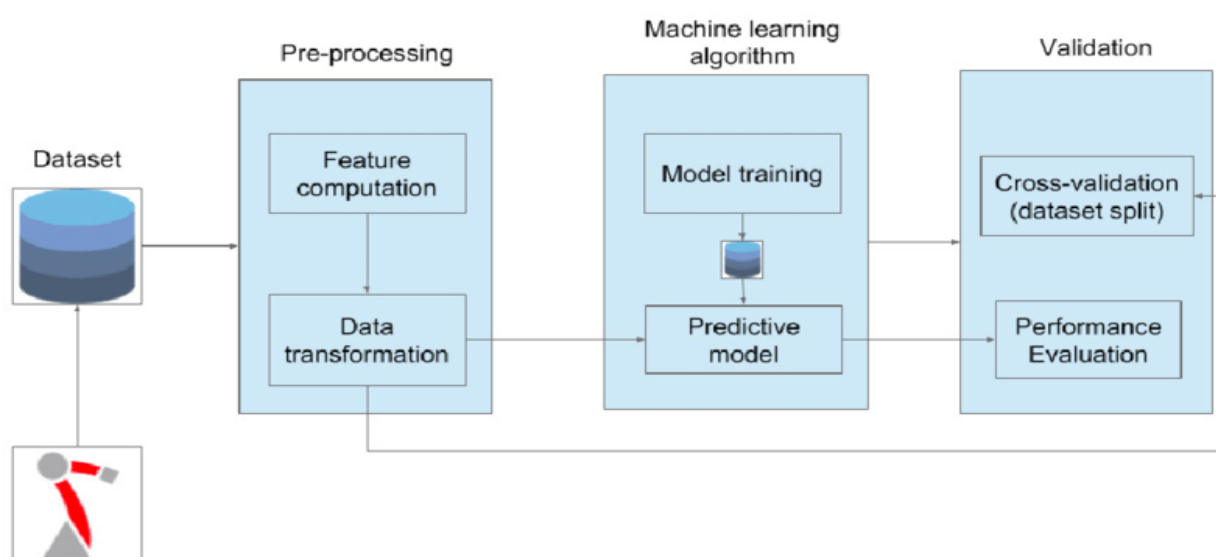
2.1 Overview of the Architecture

The architecture of the HR Analytics project is designed to support the systematic analysis of employee data to predict attrition. It consists of several interconnected components that work together to process, analyze, and model the data.

The architecture follows a modular design, allowing for flexibility, scalability, and maintainability. The main components include:

- **Data Collection Module**
- **Data Preprocessing Module**
- **Exploratory Data Analysis (EDA) Module**
- **Feature Engineering Module**
- **Model Development Module**
- **Model Evaluation Module**
- **Reporting Module**

Each of these components plays a vital role in transforming raw employee data into valuable insights for HR decision-making.



2.2 Architectural Components

1. Data Collection Module

- **Description:** This module gathers employee data from various sources, including internal databases and external files (e.g., CSV).
- **Role:** Responsible for collecting and structuring data, ensuring all necessary attributes are available for analysis.

2. Data Preprocessing Module

- **Description:** This module cleans and prepares the raw data for further analysis.
- **Role:** Ensures data quality by handling missing values, encoding categorical variables, and normalizing continuous variables.

3. Exploratory Data Analysis (EDA) Module

- **Description:** This module performs initial analyses to understand the dataset.
- **Role:** Provides visualizations and statistics that help identify trends, relationships, and potential outliers within the data.

4. Feature Engineering Module

- **Description:** This module selects and creates relevant features for the predictive model.
- **Role:** Enhances the dataset by identifying significant variables and transforming or creating new features based on insights from EDA.

5. Model Development Module

- **Description:** This module focuses on training machine learning models to predict attrition.
- **Role:** Utilizes various algorithms (e.g., Logistic Regression, Random Forest) to build predictive models based on the selected features.

6. Model Evaluation Module

- **Description:** This module evaluates the performance of the trained models.
- **Role:** Uses metrics such as accuracy, precision, and recall to assess how well the models predict attrition.

7. Reporting Module

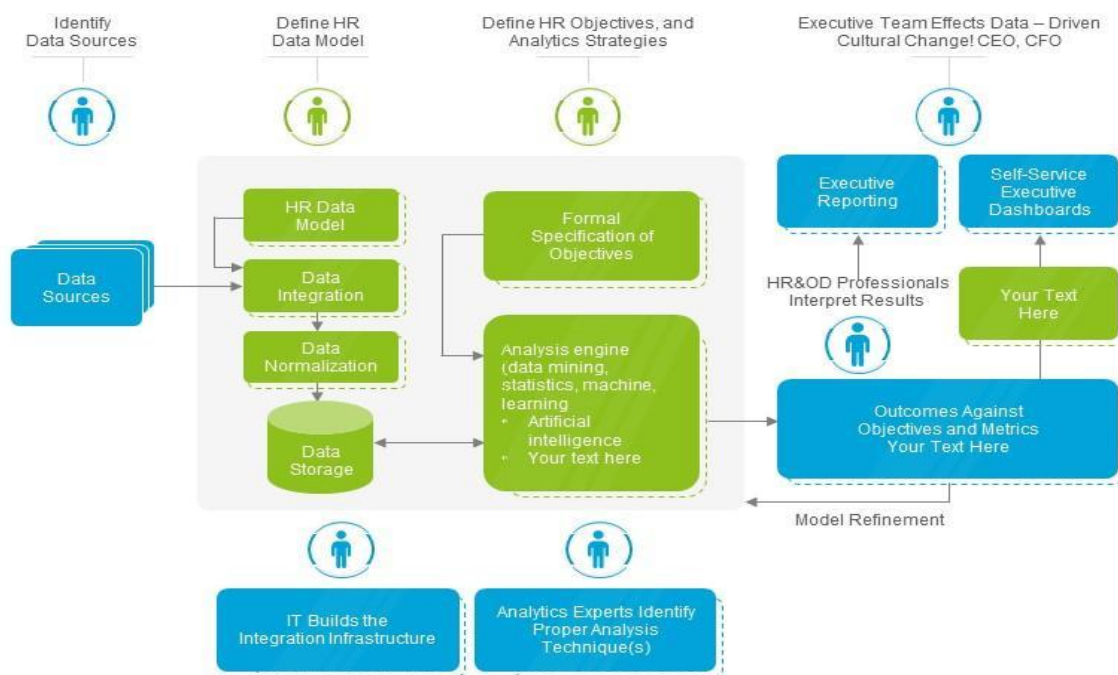
- **Description:** This module generates reports summarizing the findings from the analysis.
- **Role:** Compiles insights, visualizations, and model results into a comprehensive report for stakeholders.

2.3 Data Flow and Communication

The architecture facilitates a clear flow of data and communication between the modules:

- **Data Flow:**
 - Raw data flows from the **Data Collection Module** to the **Data Preprocessing Module**, where it is cleaned and transformed.
 - The preprocessed data is passed to the **EDA Module** for initial analysis.
 - Insights from the EDA inform the **Feature Engineering Module**, which prepares the data for modeling.
 - The **Model Development Module** uses the engineered features to train machine learning models.
 - Performance metrics from the **Model Evaluation Module** inform the final report generated by the **Reporting Module**.
- **Communication:** Each module communicates through well-defined interfaces, ensuring that data is correctly formatted and validated before being passed to the next module.

Key Components of HR Analytics Framework



3. Deployment Considerations

The deployment of the HR Analytics project involves making the various components accessible and ensuring that the insights generated can be effectively utilized.

3.1 Deployment Options

- **Local Deployment:** The project can be run on a local machine for development and testing purposes. This setup allows for quick iterations and debugging during the initial phases of the project.
- **Cloud Deployment:** For scalability, the project can be deployed on cloud platforms (e.g., AWS, Google Cloud) to handle larger datasets and provide access to a broader audience. This approach allows for better resource management and availability.

3.2 Environment Setup

- **Environment Configuration:** Ensure the deployment environment is properly configured with the necessary Python libraries (e.g., Pandas, NumPy, Scikit-learn, Matplotlib) to support data processing, analysis, and modeling.

3.3 Output Management

- **Exporting Results:** The final reports, visualizations, and model predictions should be easily exportable in user-friendly formats (HTML, PDF) for presentation to stakeholders.

3.4 Version Control

- **GitHub Repository:** The project will be maintained in a GitHub repository to enable version control, track changes, and facilitate collaboration among team members. This ensures that everyone has access to the latest code and documentation.

4. Conclusion

This Architecture Design Document for the HR Analytics project provides a detailed overview of the system's architecture and its components. Each module is designed to facilitate effective data processing and model development while ensuring clear communication between modules. The deployment strategies outlined in this document ensure that insights generated from the analysis can be effectively utilized in a real-world setting, empowering HR professionals to make informed decisions regarding employee attrition.